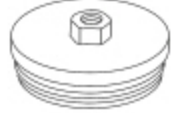


Crankshaft Rear Seal with Retainer Plate



Special Tool(s)

 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST2980-A	Installer, Rear Main Seal 303-1250
 ST1382-A	Remover, Crankshaft Rear Oil Seal 303-519 (T95P-6701-EH)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Material

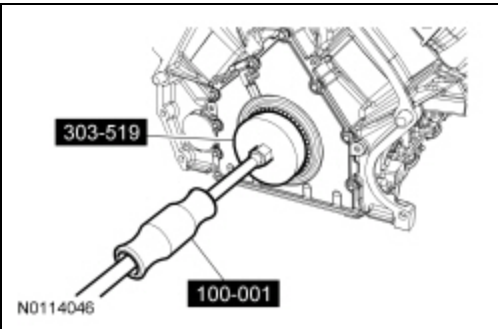
Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS- M2C945-A
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4
Motorcraft® Silicone Gasket Remover ZC-30-A	—

Removal

1. Remove the oil pan. For additional information, refer to Oil Pan in this section.
2. Remove the flexplate. For additional information, refer to Flexplate in this section.
3. Remove the engine rear cover plate.
4. **NOTE:** *Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be discarded.*

Remove the crankshaft ignition pulse ring.

5. Disconnect the Crankshaft Position (CKP) sensor electrical connector.
6. Remove the bolt and the CKP sensor.
7. Using the Slide Hammer and the Crankshaft Rear Oil Seal Remover, remove and discard the crankshaft rear seal.



8. Remove the 8 bolts and the crankshaft rear seal retainer plate.

Installation

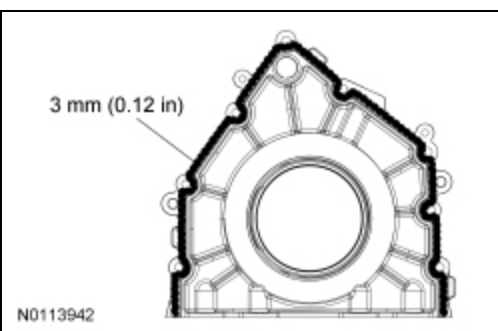
1. **NOTICE:** **Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.**

NOTE: *Clean the sealing surfaces with silicone gasket remover and metal surface prep. Follow the directions on the packaging.*

Clean and inspect the mating surface.

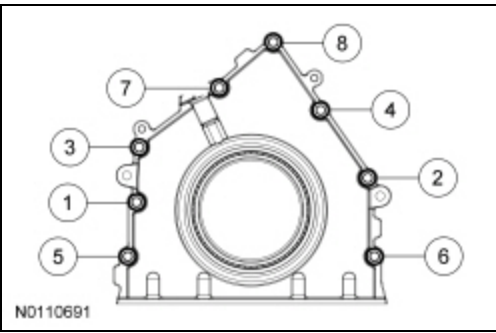
2. **NOTE:** *If the rear crankshaft seal retaining plate is not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Follow the directions on the packaging. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure may cause future oil leaks.*

Apply a bead of silicone gasket and sealant around the groove along the rear of crankshaft rear seal retainer plate sealing surface.



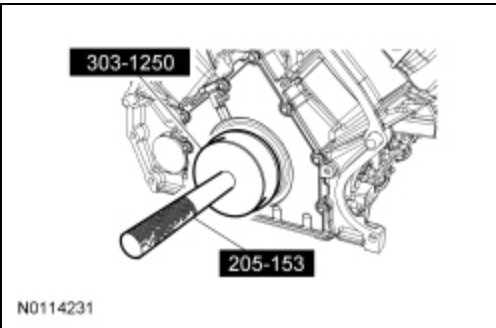
3. Install the crankshaft rear seal retainer plate and the 8 bolts in the sequence shown, in 2 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



4. **NOTE:** Lubricate the crankshaft rear seal with clean engine oil prior to installation.

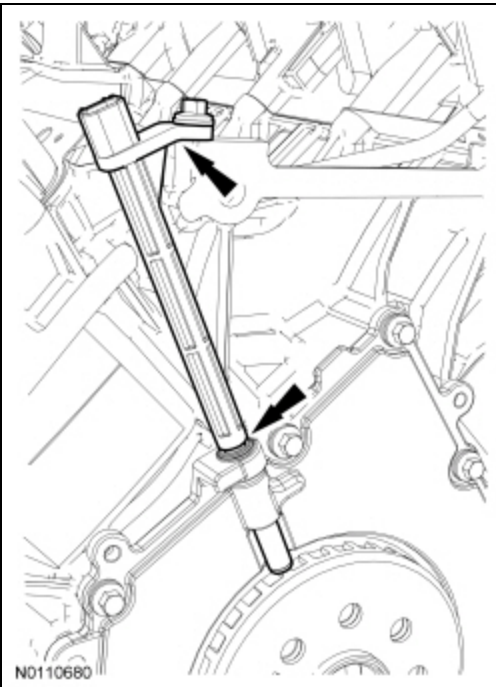
Using the Rear Main Seal Installer and Handle, install a new crankshaft rear seal.



5. **NOTICE:** The Crankshaft Position (CKP) sensor must be positioned into the fitting on the crankshaft rear seal retainer plate and be flush against the boss on the engine block before the bolt is installed. If the CKP sensor is installed incorrectly, the CKP sensor can be damaged.

Position the CKP sensor and install the bolt.

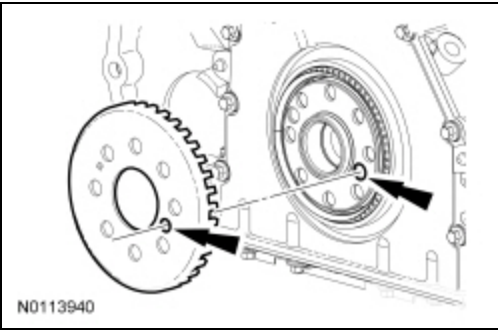
- Tighten to 10 Nm (89 lb-in).



6. Connect the CKP sensor electrical connector.

7. **NOTE:** Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be discarded.

With the inset hole on the crankshaft ignition pulse ring aligned with the inset hole on the crankshaft flange, install the crankshaft ignition pulse ring.



8. Install the engine rear cover plate.
9. Install the flexplate. For additional information, refer to Flexplate in this section.
10. Install the oil pan. For additional information, refer to Oil Pan in this section.
11. Using the scan tool, perform the Misfire Monitor Neutral Profile Correction procedure, following the on-screen instructions.